

Cognifyz Technologies Data Analysis Internship Report

Submitted By

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Domain: Data Analysis Internship

Organization: Cognifyz Technologies

Introduction

This project was completed as part of the Data Analysis Internship at Cognifyz Technologies. The objective of this internship project was to perform exploratory data analysis on a restaurant dataset using Python libraries such as Pandas and Matplotlib.

The project involved analyzing restaurant data to identify cuisine trends, city-wise restaurant distribution, price range analysis, online delivery insights, restaurant ratings, cuisine combinations, geographic patterns, and restaurant chain analysis.

The analysis was performed using Jupyter Notebook.

Tools and Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - Matplotlib
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Dataset Information

The dataset contains information about restaurants including:

- Restaurant Name
- City
- Cuisines
- Aggregate Rating
- Votes
- Price Range
- Online Delivery

- Latitude
- Longitude

The dataset was analyzed using data cleaning and visualization techniques.

LEVEL 1 TASKS

Task 1: Top Cuisines

Objective

Determine the top three most common cuisines in the dataset and calculate the percentage of restaurants serving each cuisine.

Process

- Loaded the dataset using Pandas.
- Extracted cuisines from the "Cuisines" column.
- Split multiple cuisines using commas.
- Counted cuisine occurrences.
- Identified top 3 cuisines.
- Calculated percentages.
- Created a bar chart visualization.

Result

The analysis identified the most popular cuisines available in the dataset.

Libraries Used

- Pandas
 - Matplotlib
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Task 2: City Analysis

Objective

- Identify the city with the highest number of restaurants.
- Calculate average ratings for restaurants in each city.
- Determine the city with the highest average rating.

Process

- Used `value_counts()` to count restaurants by city.
- Used `groupby()` to calculate average ratings.
- Visualized top cities using a bar chart.

Result

The analysis showed the city with the largest restaurant presence and identified cities with high average restaurant ratings.

Libraries Used

- Pandas
 - Matplotlib
-

Task 3: Price Range Distribution

Objective

- Analyze the distribution of restaurant price ranges.
- Calculate percentage distribution for each price range category.

Process

- Counted restaurants based on price range.
- Calculated percentages.
- Created a bar chart visualization.

Result

The analysis provided insights into the affordability and pricing trends of restaurants.

Libraries Used

- Pandas
 - Matplotlib
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Task 4: Online Delivery

Objective

- Determine the percentage of restaurants offering online delivery.
- Compare ratings of restaurants with and without online delivery.

Process

- Counted online delivery availability.
- Calculated percentages.
- Compared average ratings using groupby().
- Visualized distribution using a bar chart.

Result

The analysis showed the relationship between online delivery services and restaurant ratings.

Libraries Used

- Pandas
 - Matplotlib
-

LEVEL 2 TASKS

Task 1: Restaurant Ratings

Objective

- Analyze the distribution of aggregate ratings.
- Determine the most common rating range.
- Calculate average number of votes received.

Process

- Analyzed rating distribution.
- Created histogram visualization.
- Calculated average votes using mean().

Result

The analysis identified common rating patterns and customer engagement through votes.

Libraries Used

- Pandas
 - Matplotlib
-

Task 2: Cuisine Combination

Objective

- Identify common cuisine combinations.
- Determine cuisine combinations with higher ratings.

Process

- Counted cuisine combinations.
- Calculated average ratings for cuisine combinations.
- Visualized top combinations using bar charts.

Result

The analysis revealed popular cuisine pairings and their impact on ratings.

Libraries Used

- Pandas
 - Matplotlib
-

Task 3: Geographic Analysis

Objective

- Plot restaurant locations using latitude and longitude.
- Identify restaurant clusters and geographic patterns.

Process

- Used scatter plot visualization.
- Analyzed restaurant coordinates.

Result

The analysis showed concentration of restaurants in certain geographic areas.

Libraries Used

- Pandas
 - Matplotlib
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Task 4: Restaurant Chains

Objective

- Identify restaurant chains in the dataset.
- Analyze ratings and popularity of restaurant chains.

Process

- Counted repeated restaurant names.
- Identified chains using `value_counts()`.
- Calculated average ratings for chains.
- Visualized top chains using bar charts.

Result

The analysis identified popular restaurant chains and compared their ratings.

Libraries Used

- Pandas
 - Matplotlib
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Conclusion

This internship project provided practical experience in data analysis, data visualization, and exploratory data analysis using Python.

Through this project, important insights were obtained regarding restaurant trends, customer preferences, pricing, cuisine popularity, online delivery services, ratings, geographic distribution, and restaurant chains.

The project also improved skills in:

- Data Cleaning
- Data Visualization
- Python Programming
- Exploratory Data Analysis
- Problem Solving
- Working with Real-World Datasets

This internship experience helped strengthen understanding of data analysis concepts and practical implementation using Python tools.